PEGASUS
Final Presentation

Team Members:
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Advised by faculty member Dr. Knisley

Outline

- Project Goals and Overview
- Background
  - Fundamentals of operation
- Methods
  - Design and Assembly
  - Engine/Reformer Operation
- Discussion of Results
- What's next?
- Questions

Project Goals

- Design and construct a prototype
  - Capable of operating on various non-petroleum fuels
  - Increase efficiency of fuel
  - Ability to be modified and tested

- Explain, in detail, operation of reformer
  - Reformation fundamentals
  - Schematics and Prototype
  - Written Operation Procedure
  - Photos and Video clips

Project Overview

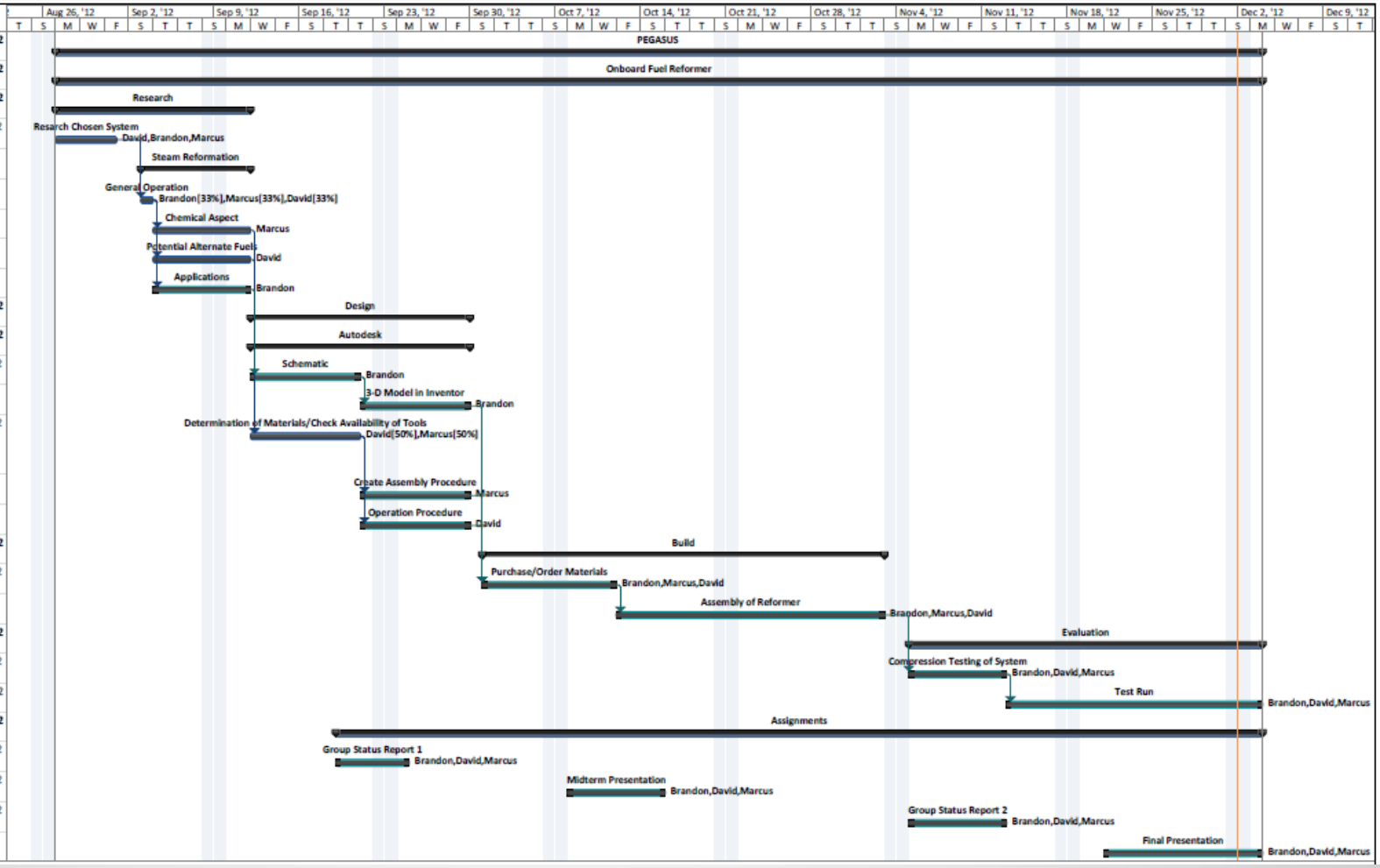
- Separated into 3 main phases
  - Research
  - Design and Construction
  - Testing

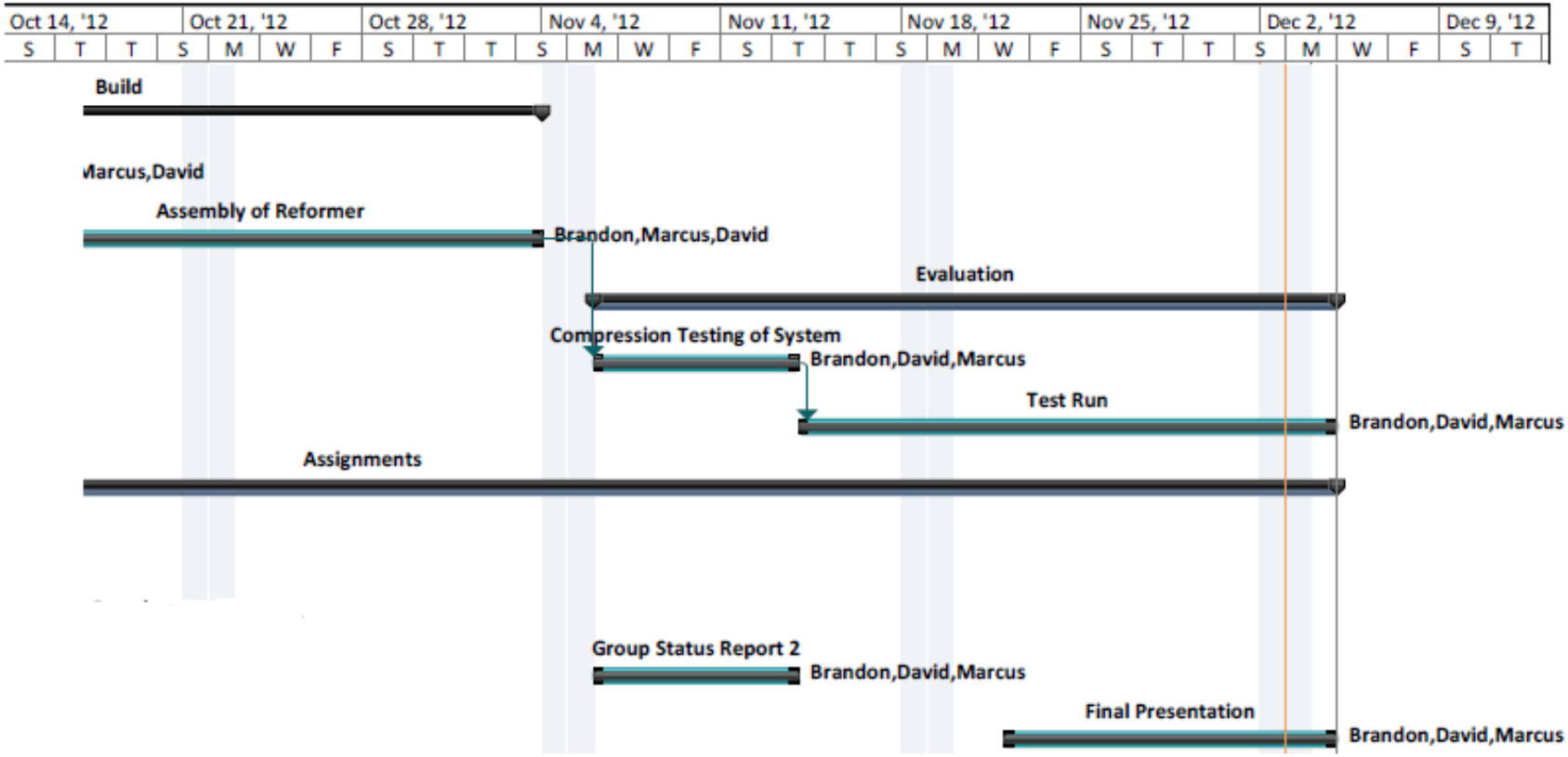
- Majority of tasks completed on time
  - Alteration of design
  - Replacement of failed parts

- Maintained below budget of $500
  - Total Parts: $272.36
  - Lawnmower: $156.45

Gantt Chart

ID	Task Name	Duration	Start
1	**PEGASUS**	**72 days**	**Mon 8/27/12**
2	**Onboard Fuel Reformer**	**72 days**	**Mon 8/27/12**
3	**Research**	**12 days**	**Mon 8/27/12**
4	Resarch Chosen System	5 days	Mon 8/27/12
5	**Steam Reformation**	**7 days**	**Mon 9/3/12**
6	General Operation	1 day	Mon 9/3/12
7	Chemical Aspect	6 days	Tue 9/4/12
8	Potential Alternate Fuels	6 days	Tue 9/4/12
9	Applications	6 days	Tue 9/4/12
10	**Design**	**14 days**	**Wed 9/12/12**
11	**Autodesk**	**14 days**	**Wed 9/12/12**
12	Schematic	7 days	Wed 9/12/12
13	3-D Model in Inventor	7 days	Fri 9/21/12
14	Determination of Materials/Check Availability of Tools	7 days	Wed 9/12/12
15	Create Assembly Procedure	7 days	Fri 9/21/12
16	Operation Procedure	7 days	Fri 9/21/12
17	**Build**	**25 days**	**Mon 10/1/12**
18	Purchase/Order Materials	9 days	Mon 10/1/12
19	Assembly of Reformer	16 days	Fri 10/12/12
20	**Evaluation**	**21 days**	**Mon 11/5/12**
21	Compression Testing of System	6 days	Mon 11/5/12
22	Test Run	15 days	Tue 11/13/12
23	**Assignments**	**55 days**	**Wed 9/19/12**
24	Group Status Report 1	4 days	Wed 9/19/12
25	Midterm Presentation	6 days	Mon 10/8/12
26	Group Status Report 2	6 days	Mon 11/5/12
27	Final Presentation	9 days	Wed 11/21/12




Oct 14, '12
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Nov 4, '12
Nov 11, '12
Nov 18, '12
Nov 25, '12
Dec 2, '12
Dec 9, '12
S T T S M W F S T T S M W F S T T S M W F S T T S M W F S T
Build
Marcus,David
Assembly of Reformer
Brandon,Marcus,David
Evaluation
Compression Testing of System
Brandon,David,Marcus
Test Run
Brandon,David,Marcus
Assignments
Group Status Report 2
Brandon,David,Marcus
Final Presentation
Brandon,David,Marcus

Project Background

- Reformation creates hydrogen syngas
  - Power fuel cells
  - Reduce engine emissions

- Methane reformation for hydrogen production
  - Used on industrial scale

- Reformation using exhaust
  - Endothermic Reaction
  - $CH_3OH + 0.5(CO_2 + 2H_2O + 5.64N_2)$ ----> $1.5CO_2 + 3H_2 + 2.82N_2$

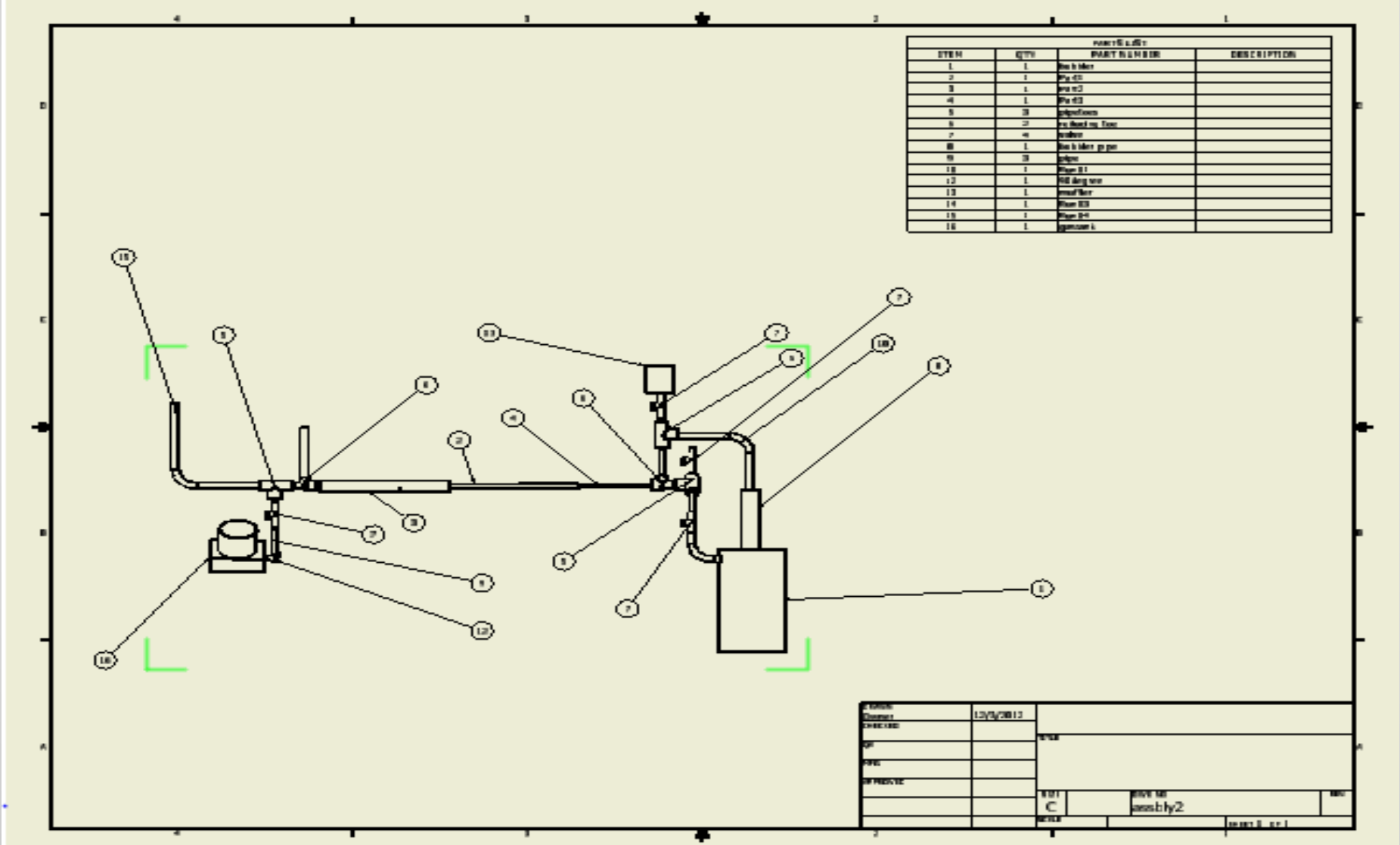
Methods
Design

- Reformer Design
  - Original plan
  - Unexpected limitations
  - Final Design

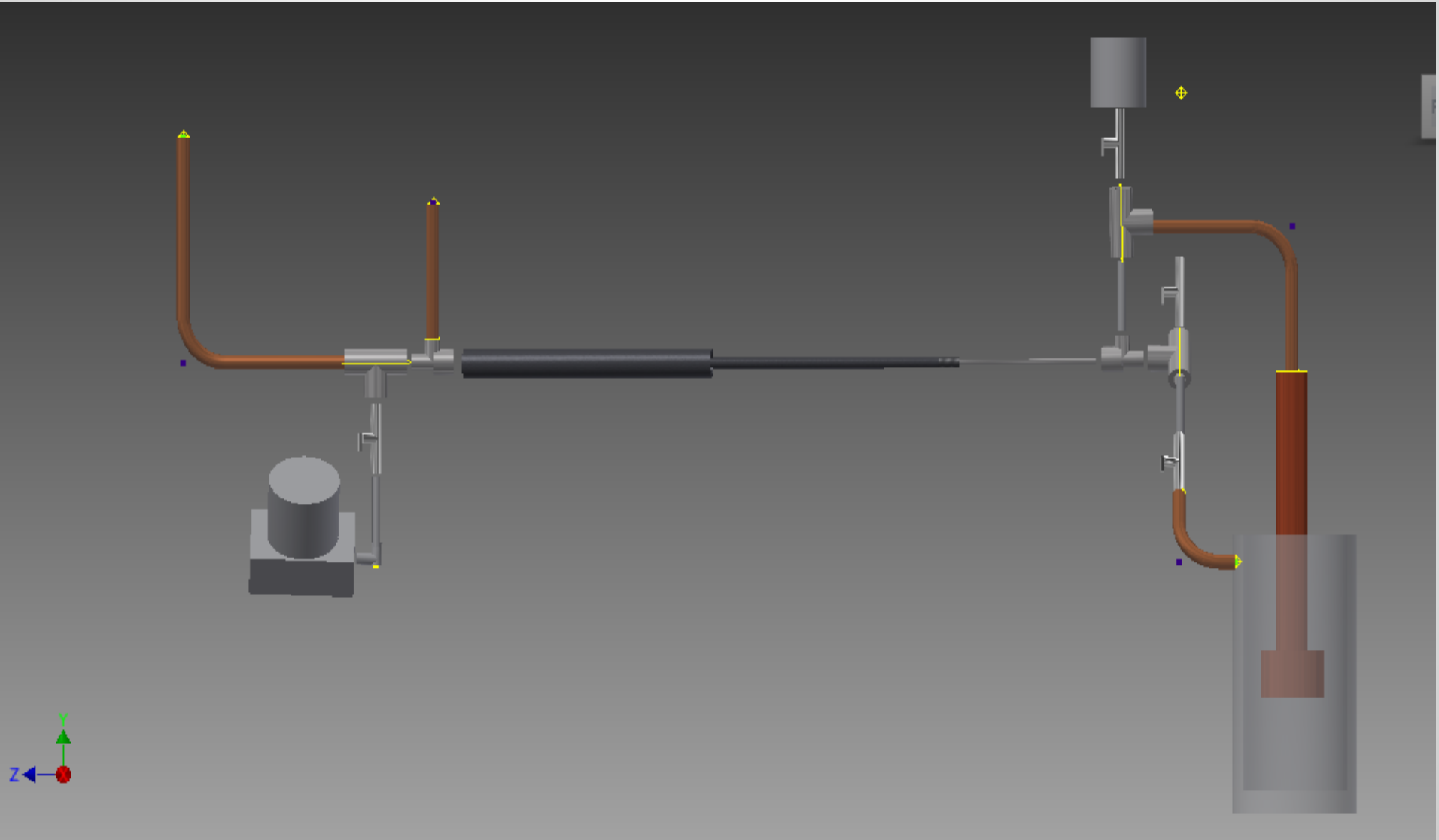
Methods
Assembly

- Reaction Chamber
- Exhaust apparatus
- Gasoline tank bypass
- Bubbler

Schematics



3-D Model



Methods
Reformer Operation

- Gasoline operation from stock tank and carburetor
  - Starting fuel to heat reformer
  - Operates exactly as normal engine

- Alternate fuels operated from bubbler
  - Exhaust gases vaporize fuel in bubbler
  - Incoming fuel vapor reformed by exhaust gas

WARNING
DANGER

MTD
Side Discharge
DANGER
WARNING

Methods
Reformer Operation (Cont)

- Fuel/Air mixture controlled through throttle valves
  - Air intake
  - Exhaust backpressure
  - Fuel intake

- Operation Video

http://www.youtube.com/watch?v=N2E2fk6rXXQ&feature=plcp

Results and Discussion

- Engine/reformer operated using non-petroleum fuels and unconventional petroleum mixtures
  - Ethanol
  - Methanol
  - Alcohol/Diesel Mixture
  - Gasoline/Alcohol Mixture
  - Gasoline/Alcohol/Diesel Mixture

- Design allowed for operator to control engine operation
  - Air/fuel ratios tested

Efficiency

Liquid Fuel	Equivelant Tank of Gasoline w/o System	Equivalent Tank of Gasoline w/ System	Efficiency Increase
Gasoline	1	0.8	20%
Ethanol	1.5	1.25	16.67%
Methanol	1.8	1.55	13.88%
E85	1.4	1	28.57%
Diesel	N/A	Will now operate a gasoline engine	N/A

Results and Discussion (Cont)

- Reformer operation and fundamentals described in final report
  - Flow path of exhaust and fuels
  - Chemical operation of reformation
  - Reformer assembly procedure and parts list

What's Next?
Future of PEGASUS Project

- Testing using additional various fuels
- Engine output
  - Tachometer
  - Dynamometer
- Emissions testing

Questions?
MTD
Side Discharge
WARNING